

Phase Noise Effects on Micro-Doppler Extraction for Radar Target Classification using Machine Learning

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Abstract—The extraction of Micro-Doppler (M-D) signatures from radar signals is critical for classifying various targets. This study uses Machine Learning techniques to investigate the impact of phase noise on micro-Doppler extraction.

Experiments are conducted with varying phase noise intensities to assess its influence on classification performance. Detailed micro-Doppler simulations for different objects illustrate how phase noise impacts their distinctive signatures. Using the micro-Doppler signal-to-noise ratio (MSNR), results show that increasing carrier frequency does not enhance the extraction performance of micro-motion features.

Index Terms—Machine Learning, RADAR, Target Classification, Phase noise, Direct RF transceiver, Convolutional Neural Network (CNN), Micro Doppler

I. INTRODUCTION

Accurate detection and classification of radar targets are critical for various defense, aviation, and autonomous systems applications. One of the significant factors affecting radar performance is phase noise, which manifests as additional noise in the system, impacting the coherent integration of pulses and the ability to resolve targets. This study focuses on the impact of phase noise on the signal-to-noise ratio (SNR) and the extraction performance of micro-motion features in radar systems.

Phase noise is often characterized by its power spectral density (PSD), which describes the power distribution of the phase fluctuations over frequency. The single-sideband (SSB) phase noise at an offset frequency from the carrier frequency is a key metric, as it contributes to the noise floor, effectively reducing the SNR. Understanding and modeling phase noise is essential for accurately predicting radar performance and enhancing target detection capabilities.

In this paper, a mathematical model of phase noise is developed, and its effects on the radar equation and SNR are explored. Simulations are presented to demonstrate how phase noise influences the micro-Doppler signatures of various radar targets, such as vibrating and rotating objects, and drones. The micro-Doppler effect, which arises from the relative motion of the target's scatterers, provides valuable information for target classification and identification.

By examining the relationship between phase noise and SNR, this study aims to establish guidelines for optimizing

radar system parameters to improve detection performance. Additionally, the required SNR for different detection probabilities is calculated, and the impact of phase noise on micro-Doppler-based target classification is assessed. These findings contribute to ongoing efforts to enhance radar detection and classification accuracy in the presence of phase noise.

Zihao Liu [2] presented the results for phase noise influence on micro-Doppler feature extraction of vibrating targets. Still, he did not discuss the effect of phase noise on SNR or for multi-vibrating targets like helicopters, drones, etc.

II. CONCEPTUAL DESIGN

A. Single-sideband Phase Noise

Phase noise is often characterized by its PSD, $\mathcal{L}(f)$, which describes the power distribution of the phase fluctuations over frequency. Phase noise $\mathcal{L}(f)$ manifests as additional noise in the system, particularly affecting the coherent integration of pulses and the ability to resolve targets. It contributes to the noise floor, effectively reducing the SNR. The single-sideband (SSB) phase noise at an offset frequency f from the carrier frequency ω_0 is defined as [6]:

$$\mathcal{L}(f) = 10 \log_{10} \left(\frac{P_{\text{sideband}}(f)}{P_{\text{carrier}}} \right) \quad (1)$$

where: $P_{\text{sideband}}(f)$ is the noise power in a 1 Hz bandwidth at an offset frequency f from the carrier. P_{carrier} is the total power of the carrier signal.

B. Impact of SSB phase Noise on Radar equation and SNR

The SSB phase noise $\mathcal{L}(f)$ affects radar system performance by influencing the signal-to-noise ratio (SNR) and thereby impacting the overall detection capability of the radar.

The basic RADAR equation is given by [5]:

$$P_r = \frac{P_t G_t G_r \lambda^2 \sigma}{(4\pi)^3 R^4 L_s} \quad (2)$$

Considering the phase noise, the modified SNR can be expressed as:

$$\text{SNR} = \frac{P_r}{P_n + P_{\text{phase}}} \quad (3)$$

where P_{phase} is the power contributed by the phase noise. By using the following RADAR parameter and for different

phase noise values, the SNR has been calculated and its plot against phase noise is shown in Fig 1. $P_t = 1 \text{ W}$, $G_t = 1$, $G_r = 1$, $\lambda = 0.03 \text{ m}$, $\sigma = 10 \text{ m}^2$ (Radar cross-section), $R = 100 \text{ m}$, $L_s = 1$ (System losses)

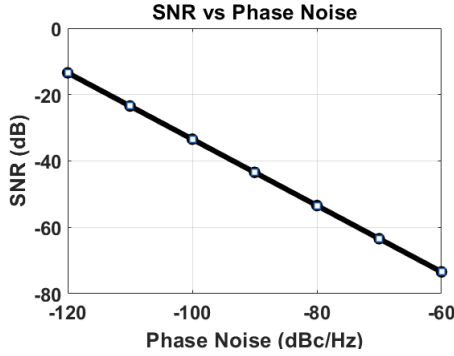


Fig. 1: SNR vs phase noise simulation plot

As seen in Fig 1, phase noise significantly impacts the received signal SNR. Simulations have also been conducted for the radar SNR required for different probabilities of detection, which will be discussed in the next section II-C.

C. SNR requirement for RADAR target detection

Phase noise directly affects the SNR requirement for radar systems. In research papers, the minimum SNR requirement for target detection is first calculated, and then multiple simulations are performed to show the effect of phase noise on micro-Doppler.

Albersheim's [3] and Shnidman's [4] equations both aim to relate key radar performance parameters such as the probability of detection (P_d), probability of false alarm (P_{fa}), SNR, and the number of pulses (N). These equations are used to calculate SNR under different radar parameter conditions.

Albersheim's Equation [3]:

$$\log_{10} \left(\frac{1}{P_d} \right) = 0.12 \left(\log_{10} \left(\frac{1}{P_{fa}} \right) \right)^{0.8} + 1.7 \quad (4)$$

Approximate SNR :

$$\text{SNR} \approx \frac{-\ln(P_{fa})}{0.12 \left(\ln \left(\frac{1}{P_d} \right) \right)^{1.25} - 1.7} \quad (5)$$

Shnidman's Equation [4]:

$$P_d = Q \left(Q^{-1}(P_{fa}) - \sqrt{2N \cdot \text{SNR}} \right) \quad (6)$$

Using equation 4, SNR vs probability of false alarm (P_{fa}) is plotted as shown in Fig 2. In the case of pulse radar, the required single-pulse SNR using equation 6 for Swerling cases I and III [7] is as shown in Fig 3.

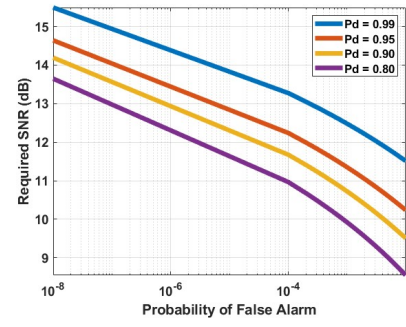


Fig. 2: Required SNR for different PD and No of pulses = 1

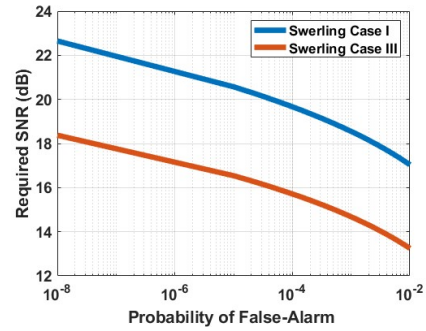


Fig. 3: Required Single-Pulse SNR for Pd = 0.9

III. SIMULATION OF MICRO-DOPPLER FROM DIFFERENT RADAR TARGETS

The previous section explained the effect of phase noise on SNR. This section presents simulations of micro-Doppler effects from vibrating targets, rotating targets, and drones. Finally, the impact of phase noise on micro-Doppler-based target classification is discussed.

A. Micro-Doppler modulation of vibrating and rotating targets

Modulation of a vibrating and rotating target is given by [1]:

$$f_{\text{micro-Doppler}} = \frac{2f_c}{c} [\hat{\omega} (\hat{\omega} \sin(\|\omega\|t) + \mathbf{I} \cos(\|\omega\|t)) \mathbf{R}_{\text{init}} \mathbf{r}_0]_{\text{radial}} \quad (7)$$

where: $\hat{\omega} = \frac{\omega}{\|\omega\|}$ is the unit velocity vector, $\mathbf{R}_{\text{init}}$ is the initial rotation matrix, $\mathbf{r}_0$ is the initial position of the target, $[\cdot]_{\text{radial}}$ indicates the radial component.

B. Micro Doppler generation for drone

Mathematical simulation for a drone having 4 rotors with 2 blades each has been done for the micro-doppler as shown in Fig 5 and the following parameters were considered for the simulation

No. of motors = 4, NO. OF BLADES = 2, SNR = 20; NFFT = 144, NO OF RETURNS = 400, BODY TO BLADE RCS RATIO = 10, distance between motors = 100m, $f_c = 6\text{e}9$, $f_s = 1\text{e}6$; $\text{prf} = 2\text{e}4$.

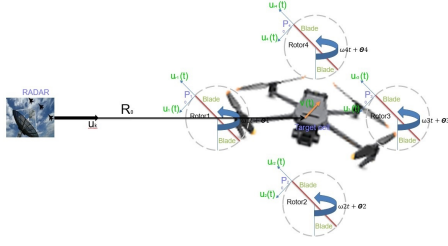


Fig. 4: RADAR setup of 4 rotors having 2 blades each

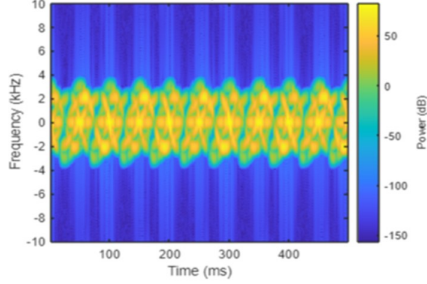


Fig. 5: Micro-Doppler plot of 4 rotors having 2 blades each

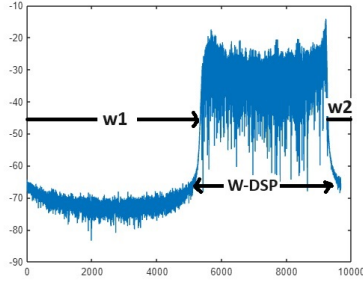


Fig. 6: FFT plot of M-D at -90 dBc phase noise

IV. RESULTS

For the effect of phase noise, the micro-Doppler of a helicopter has been simulated. Since the focus is on the Micro-Doppler signal-to-noise ratio (MSNR), the Doppler effect from the main body of the helicopter has been removed, leaving only the micro-Doppler plot as shown in Fig 6. The MSNR has been calculated by averaging the left and right noise windows from the micro-Doppler signal.

The noise power can be calculated using the formula

$$\text{Noise Power(Nd)} = \frac{w_1 + w_2}{2} \quad (8)$$

w_1 and w_2 are left and right windows as shown in Fig 6

Micro Doppler signal power (M-DSP) is calculated using the central window as shown in Fig 6. So MSNR can be

$$\text{MSNR} = \frac{\text{Micro Doppler Signal Power (M-DSP)}}{\text{Noise Power(Nd)}} \quad (9)$$

A. MSNR calculation for different phase noise at 100Hz offset

Using equation 9, micro-doppler signal power is -31.1351 dBm and Noise power is -70.6449 dBm. So MSNR is 39.50 dB. Similarly, MSNR has been calculated for different phase noise values and carrier frequencies as shown in Fig 7

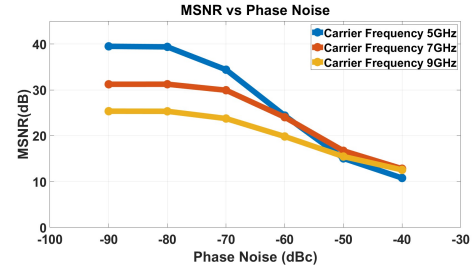


Fig. 7: MSNR Vs phase noise at different carrier frequencies

According to mathematical simulations, a target is detected if its MSNR exceeds 12 dB. As shown in Fig 7, at a 5 GHz carrier frequency, the MSNR is 12 dB for phase noise of -40 dBc/Hz. Therefore, the target cannot be detected if the phase noise is less than -40 dBc/Hz at 5 GHz.

Convolutional Neural Network (CNN) based machine learning model has been used for target classification against difference phase noise values. Model accuracy at different phase noise levels is shown in Fig 8.

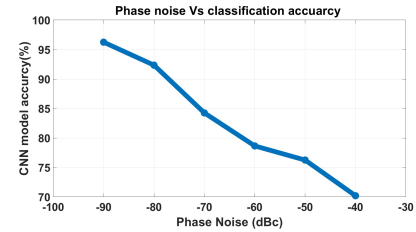


Fig. 8: Phase noise Vs classification accuracy

V. CONCLUSION

The study demonstrates the significant impact of phase noise on micro-Doppler feature extraction and target classification in radar systems. Experiments conducted with varying phase noise intensities reveal that degraded phase noise degrades micro-Doppler signal quality, reducing classification accuracy. Simulation results show that higher carrier frequencies do not mitigate phase noise effects. The proposed MSNR metric effectively quantifies phase noise impact, with detection thresholds established through mathematical simulations. The findings emphasize the need for optimizing radar system parameters to manage phase noise and enhance target detection and classification accuracy, contributing to improved radar performance in noisy environments.

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